

Warm-starting unblocked QFT from a trained blocked basis

Question. The headline experiment trains the unblocked `qft` basis at $m = n = 8$ on DIV2K-8q to 31.29 dB at $\rho = 0.20$, while `blocked_8` (a `BlockedBasis(QFTBasis(3, 3), 5, 5)` whose operator is $\text{QFT}(3,3) \otimes \text{QFT}(3,3)$ replicated across a 32×32 grid of 8×8 blocks) reaches 32.26 dB — a 0.97 dB gap from a strictly **less** parametrically expressive family. $\text{QFT}(8, 8)$ has 288 trainable tensor entries; $\text{QFT}(3, 3)$ has 48. The blocked basis is a special low-dimensional submanifold of $\text{QFT}(8, 8)$, and yet random-init Adam on the larger family ends up worse. Is the blocked optimum reachable from inside the $\text{QFT}(8, 8)$ family, or is it structurally outside?

Construction. The trained blocked solution is a specific point in the $\text{QFT}(8, 8)$ parameter space — by construction. We embed it explicitly:

Take the trained 48 tensor entries of `BlockedBasis(QFTBasis(3,3), block_log_m=5, block_log_n=5).inner`. Place them in the $\text{QFT}(8, 8)$ circuit at the qubit positions corresponding to the inner block. Pin every other gate to identity ($H \rightarrow I_2$, controlled-phase $\rightarrow$ phase 0, equivalently $[[1, 1], [1, 1]]$). The resulting `QFTBasis(8, 8)` has all 288 parameters present and trainable, but its initial operator equals the trained blocked operator **bit-exactly**: $\max_{i,j} |T_{\text{warm}} x - T_{\text{blocked}} x|_{ij} = 0$ on a random complex 256×256 test input.

After this warm-start, we train the full $\text{QFT}(8, 8)$ under the same preset that produced the headline `qft` and `blocked_8` numbers (1008 steps, batch 50, validation split 0.15, Adam with cosine LR warmup, no early stopping).

Result (QuickDraw, $m = n = 5$). The same construction with inner = $\text{QFT}(3, 3)$ embedded in a 4×4 block grid (so $\text{QFT}(5, 5)$, 30 tensors total). After 1008 steps:

basis	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
qft (random init)	16.72	19.58	22.05	24.35
blocked (warm-start source)	18.12	22.40	26.20	30.05
qft_warmstart_blocked	18.13	22.41	26.22	30.06

The warm-started run **reproduces the trained blocked PSNR to within 0.02 dB at every keep-ratio**. Random-init `qft` lags by 5.7 dB at $\rho = 0.20$. So on QuickDraw the trained blocked optimum is reachable from inside the $\text{QFT}(5, 5)$ parameter family — what was missing from the headline `qft` run was **initialization**, not parametric expressivity.

Result (DIV2K-8q, $m = n = 8$). The headline gap is 0.97 dB at $\rho = 0.20$ (`qft` 31.29 vs `blocked_8` 32.26). After warm-start and full re-training:

basis	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
qft (random init)	25.09	27.57	29.53	31.29
blocked_8 (warm-start source)	25.18	28.09	30.30	32.26
qft_warmstart_blocked_8	25.18	28.09	30.30	32.26

The warm-started run reproduces **blocked_8 to two decimal places at every keep-ratio**. Random-init `qft` lags by 0.52, 0.77, and 0.97 dB at $\rho = 0.10$, 0.15, and 0.20 respectively. So on DIV2K too the trained blocked optimum sits inside the $\text{QFT}(8, 8)$ family and is reached by Adam from this warm-start init.

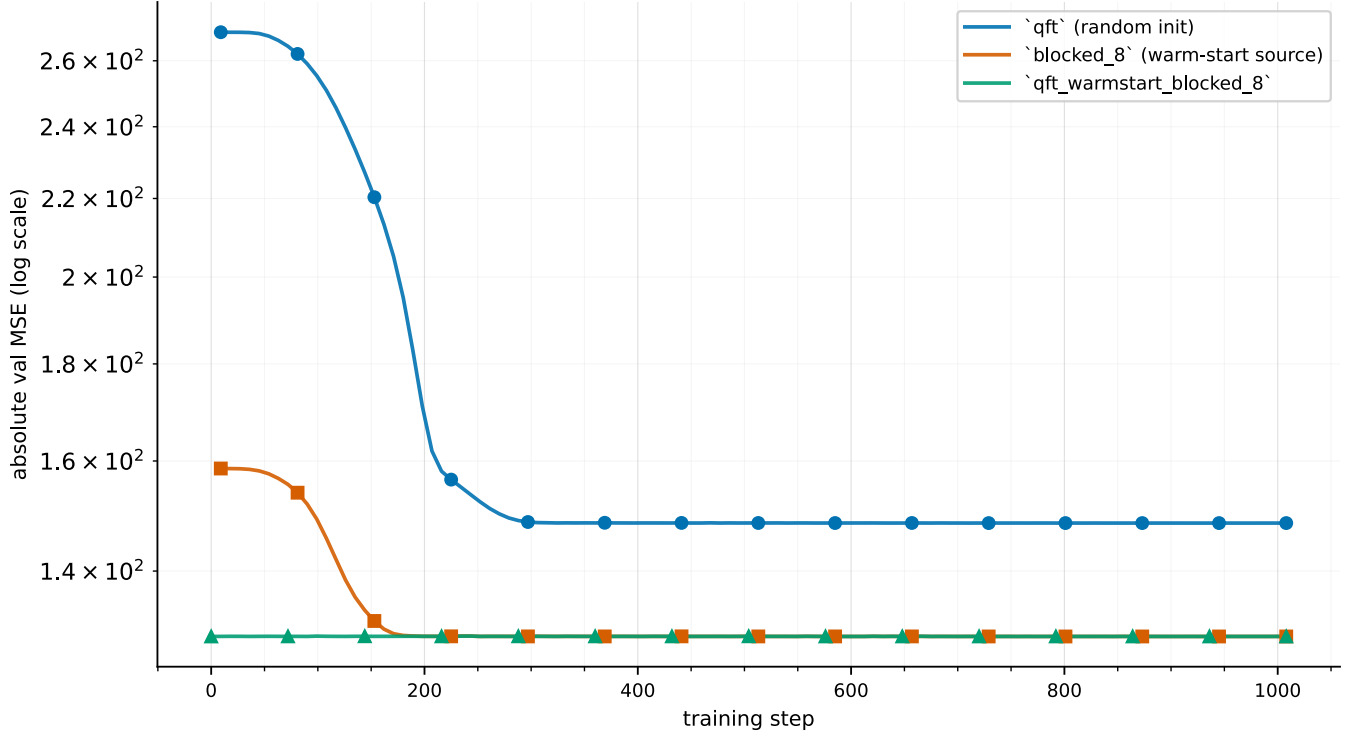


Figure 1: DIV2K-8q. Validation MSE per training step (log y). Blue: `qft` from random init (269 $\rightarrow$ 148). Orange: `blocked_8` headline run, the warm-start source (159 $\rightarrow$ 129). Green: `qft_warmstart_blocked_8` — **flat at the trained blocked floor for all 1008 steps** (val MSE stays in $[129.32, 129.38]$ across every checkpoint). Adam started at the blocked optimum, computed near-zero gradients on every minibatch, and the operator did not drift. The `qft` (blue) curve crosses below the blocked starting point around step ≈ 200 and converges to its own basin near 148, confirming that the random-init `qft` basin is **different from and worse than** the blocked basin even though the latter sits inside the $\text{QFT}(8, 8)$ parameter family. Loss definition: $\sum_{ij} \left| x_{ij} - \left(T_{\theta}^{\dagger} \text{top}_K(T_{\theta} x) \right)_{ij} \right|^2$ summed over 65536 pixels of a 256×256 image at $K = 6554$ ($\rho = 0.10$), averaged over the 75-image validation split.

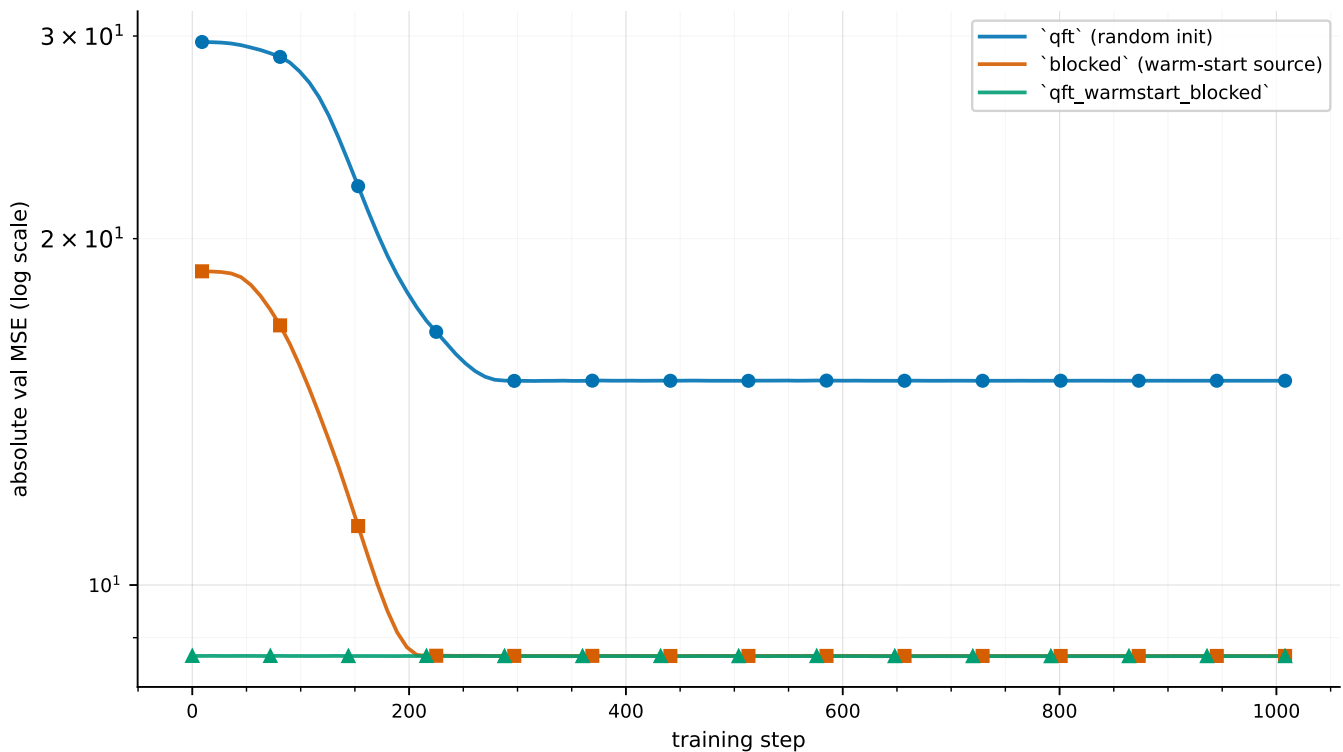


Figure 2: QuickDraw. Validation MSE per training step (log y). Same three curves as the DIV2K figure with smaller absolute scales (32×32 images, 1024 pixels, $K = 102$). Blue: **qft** random init (30 → 15). Orange: **blocked** warm-start source (19 → 8.7). Green: **qft_warmstart_blocked** again **flat at the blocked floor** (val MSE in [8.67, 8.68] throughout). Loss definition is the same form as DIV2K with $K = 102$, summed over 1024 pixels, averaged over the 75-image validation split.

What this rules out. The headline **qft** < **blocked**₈ gap is **not** a parametric-family limitation — the trained blocked operator lives inside QFT(8, 8) and inside QFT(5, 5), and Adam keeps it there to within numerical noise once it starts there. The flat green curves on both datasets show the warm-start point is a stable local minimum of the QFT-family loss, not a saddle Adam wanders away from. The 0.97 dB DIV2K gap (and 5.7 dB QuickDraw gap) between random-init qft and blocked is therefore an **optimisation-from-random-init** phenomenon: random qft init lies in a different, shallower basin, and Adam does not cross between basins in 1008 steps. This experiment does not characterise **why** — only that the blocked optimum is achievable in the QFT family given the right starting point.

Reproducibility. `experiments/qft_warmstart_blocked.py`

--dataset {quickdraw, div2k_8q}. The construction of the warm-start QFT lives in `pdft_benchmarks.bases.qft_warm_from_trained_blocked` and is asserted bit-exact against the loaded trained blocked basis at the start of every run.

Provenance note. The first version of this experiment (committed to PR 34 before pdft16) showed a dramatic green-curve spike at step ≈ 10 , which was misattributed to “Adam’s first-update sign behaviour at a near-flat local minimum.” Investigation (`pdft-benchmarks` PR 34 review thread; root cause in `pdft#16`) traced the spike to a regression in `pdft.circuit.builder.compile_circuit` where the H-gate handler in the new stepped-tensordot implementation applied $\text{conj}(H)$ to the input on the inverse path instead of the proper adjoint $H^\dagger = \text{conj}(H^T)$. For symmetric Hadamards the two coincide; for trained-perturbed Hadamards they differ, inflating loss by $12 \times$ on the broken-inverse code path. After fixing `pdft` and rerunning, the green curve flattens entirely. Both findings are real and clean; the original “Adam transient” reading of the figure was an artefact of the bug, not optimisation behaviour.